

Peiyu Zhang

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Education

University of Virginia, PhD Candidate of Systems and Information Engineering Sep 2022 – Present

- **Research Interests: Human-Computer Interaction, Artificial Intelligence, Human Factors Engineering**
- GPA: 4.0/4.0, MEng obtained: May 2025 | PhD expected: May 2027

University of Virginia, BA in Computer Science with High Distinction Sep 2020 – May 2022

- GPA: 4.0/4.0, Minor: Psychology

Technical Skills

Languages & Frameworks: Python, Java, C#, C++, SQL, Kotlin, JavaScript, PyTorch, React, Node.js

Tools & Platforms: Unity, Android Studio, AWS, MongoDB

Experience

University of Virginia Riggs' Lab, Graduate Research Assistant Jan. 2021 – Present

- Built multiple **VR/AR applications** using **Unity (C#)** and **Android Studio** for **Meta Quest**, **Microsoft HoloLens**, and **Magic Leap**, supporting **multimodal input/output testing** and interactive task-switching environments.
- Led a series of mixed reality experiments analyzing cognitive workload and multi-interface attention through **multimodal data analysis** of eye tracking, movement tracking, and event logs using **Python** and **R** for **statistical modeling and visualization**.
- Designed human-in-the-loop evaluations of AI-assisted clinical decision-making and **human-AI collaboration** workflows, and iterated on **AI system design** for diagnostic/visualization tools to enhance performance and promote calibrated trust.
- Explored **NLP/LLM** approaches for attitude/stance understanding in online discussion communities, and prototyped **LLM-agent-based** therapeutic assistance tools and VR interactive systems.
- Developed a VR application to record and **process 3D movement data**, integrating real-time sensor input and spatial tracking for **quantitative kinematic analysis**.

University of Virginia Ultimate User Interface Lab, Graduate Research Assistant Sep 2021 – Sep 2023

- Developed a **computer vision-based wearable sensing system** to detect micro-movements and physical changes (e.g., force, volume) using **Python**, **PyTorch**, **OpenCV**, and convolutional neural networks (**CNNs**) informed by the Moiré pattern optical effect.
- Built **processing pipelines** for motion tracking and trained image recognition models to detect **real-time physical interactions**, enabling tangible interfaces via spatial gesture input, and integrated the system into self-developed **Android** and **VR applications**.
- Contributed to experimental design and VR prototyping to investigate VR-based perception manipulation in reaching tasks using a **LSTM autoencoder** for **time series data analysis**.

University of Virginia Physics Department, Graduate Research Assistant May 2022 - Sep 2022

- Created an interactive VR system in Unity for immersive science education, engineering 3D models and integrating physics-based animation pipelines to visualize real-time parton dynamics.

Software Projects

AI-Powered Personal Finance Dashboard

- Developed a full-stack personal finance dashboard using **React**, **Node.js**, and **MongoDB**, allowing users to log expenses and investments, manage budgets, and track real-time stock holdings.
- Integrated live stock market data from the **Finnhub API**, enabling users to monitor stock performance, view gains and losses, and explore historical price trends.
- Built an AI-powered insights module using the **OpenAI API** to generate natural language summaries of spending habits, personalized saving tips, and stock-related news and suggestions.

- Built interactive visualizations with **Recharts** for spending, stock trends, and budget-vs-actual comparisons.

AI Agentic Travel Planner

- Developed a **LangChain-based multi-agent planner** (Planner, Verifier, Local Guide, Budgeter) to generate itineraries, validate constraints, and produce cost breakdowns.
- Built a **RAG pipeline** over personal travel notes and itineraries to ground recommendations, and created a “travel taste” **embedding profile** from past trips and saved places to rank suggestions that match user style.
- Implemented constraint-check tools and safety guardrails to reduce errors and improve trustworthiness.

Android AR Treasure Hunt Mobile Game

- Developed an AR-based Android game using **ARCore** and **Android Studio**, allowing users to explore physical environments and discover virtual collectibles overlaid in real space.
- Implemented plane detection and hit testing with ARCore to anchor virtual game objects (e.g., coins, gold bars) onto real-world surfaces in real time.
- Implemented real-time **UDP communication** for local multiplayer gameplay, allowing devices on the same network to broadcast player actions and synchronize game state.
- Designed a modular game logic system using **Kotlin**, featuring progressive difficulty, interactive animations, scorekeeping, collectible tracking, and real-time visual feedback to enhance player engagement.
- Integrated persistent game state using Room (**SQLite**) to store player progress, collected items, and scores across sessions.

Publications

Web Pages as 3D Proxies in VR: A Design Framing and Preliminary Exploration Tianjie Shu* and <i>Peiyu Zhang*</i> <i>ACM CHI EA 2026</i> , DOI: 10.1145/3772363.3798605	Apr 2026
Multitasking Across Physical and Virtual Displays: The Effect of Spatial Discontinuity and Task Load 🏆 Honorable Mention Award <i>Peiyu Zhang</i> , Mohamad El Iskandarani, and Sara L. Riggs <i>IEEE VR 2026 (Special Issue of IEEE TVCG)</i> , DOI: 10.1109/TVCG.2026.3679118	Mar 2026
The Effect of Layout on Visual Search in Augmented Reality Multi-Window Displays <i>Peiyu Zhang</i> , Mohamad El Iskandarani, and Sara L. Riggs <i>IEEE Transactions on Visualization and Computer Graphics</i> , DOI: 10.1109/TVCG.2025.3642044	Dec 2025
The Effect of AI Assistance Timing on Performance and User Perceptions in Pathological Slide Diagnosis <i>Peiyu Zhang</i> , Zichen Ye, Ronggan Wei, Haiyan Niu, Hongxia Li, Mingjuan Wang, Peng Xue, and Sara L. Riggs <i>Applied Ergonomics</i> , DOI: 10.1016/j.apergo.2025.104663	Oct 2025
Deep learning-assisted system improves practical effect in cervical cytopathology diagnosis: A comparative study of reading modes Zichen Ye, <i>Peiyu Zhang</i> , Ronggan Wei, Haiyan Niu, Hongxia Li, Mingjuan Wang, Sara L. Riggs, and Peng Xue <i>Modern Pathology</i> , DOI: 10.1016/j.modpat.2025.100882	Sep 2025
Multitasking Between Realities: An Examination of Multitasking Using Augmented Reality Under Different Task Demand Levels <i>Peiyu Zhang</i> and Sara L. Riggs <i>HFES Annual Meeting (ASPIRE) 2025</i> , DOI: 10.1177/10711813251357901	Jul 2025
MoiréTag: A Low-Cost Tag for High-Precision Tangible Interactions without Active Components 🏆 Honorable Mention Award (Top 3 Papers)	Oct 2024

Peiyu Zhang, Wen Ying, Sara L. Riggs, and Seongkook Heo

ACM ISS 2024, DOI: 10.1145/3698113

Chinese colposcopists' attitudes toward the colposcopic artificial intelligence auxiliary diagnostic system (CAIADS): A nation-wide, multi-center survey

Sep 2024

Huike Wang*, Zichen Ye*, **Peiyu Zhang***, Xiaoli Cui, Mingyang Chen, Aiyuan Wu, Sara L. Riggs, Peng Xue, and Youlin Qiao

Sage Digital Health, DOI: 10.1177/20552076241279952

Fringer: A Finger-Worn Passive Device Enabling Computer Vision Based Force Sensing Using Moiré Fringes

Oct 2022

Peiyu Zhang, Wen Ying, and Seongkook Heo

UIST 2022, DOI: 10.1145/3526114.3558706